



SOMAIYA
VIDYAVIHAR UNIVERSITY

Semester: October 2022 – January 2023 Examination: Mid Term Examination			
Programme code: 03 Programme: TY B.Tech.		Class: TY COMP	Semester: VI (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: COMP	
Course Code: 116U01C601		Name of the Course: Digital Signal and Image Processing	

Question No.		Max. Marks	CO Mapped																																				
Q1	Determine linear Convolution of following two sequences and solve using graphical method. $x(n) = \{3, 2, 1, 2\}$ $\uparrow$ $h(n) = \{1, 2, 1, 2\}$ $\uparrow$	10	CO1																																				
	OR Apply Circular Convolution on the given sequences. $x(n) = \{-1, 1, 2, -2\}$ $\uparrow$ $h(n) = \{0.5, 1, -1, 2, 0.75\}$ $\uparrow$																																						
Q2a	Apply average filters of (3 ✖ 3 mask) on the following image. <table border="1"><tr><td>1</td><td>1</td><td>2</td><td>3</td><td>2</td><td>2</td></tr><tr><td>1</td><td>1</td><td>2</td><td>3</td><td>2</td><td>2</td></tr><tr><td>4</td><td>4</td><td>2</td><td>5</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>2</td><td>6</td><td>3</td><td>3</td></tr><tr><td>2</td><td>2</td><td>4</td><td>6</td><td>7</td><td>7</td></tr><tr><td>2</td><td>2</td><td>4</td><td>6</td><td>7</td><td>7</td></tr></table>	1	1	2	3	2	2	1	1	2	3	2	2	4	4	2	5	1	1	1	1	2	6	3	3	2	2	4	6	7	7	2	2	4	6	7	7	5	CO1
1	1	2	3	2	2																																		
1	1	2	3	2	2																																		
4	4	2	5	1	1																																		
1	1	2	6	3	3																																		
2	2	4	6	7	7																																		
2	2	4	6	7	7																																		
Q2b	Apply histogram equalization on the following image <table border="1"><tr><td>2</td><td>3</td><td>3</td><td>2</td></tr><tr><td>4</td><td>2</td><td>4</td><td>3</td></tr><tr><td>3</td><td>2</td><td>3</td><td>5</td></tr><tr><td>2</td><td>4</td><td>2</td><td>4</td></tr></table>	2	3	3	2	4	2	4	3	3	2	3	5	2	4	2	4	5	CO1																				
2	3	3	2																																				
4	2	4	3																																				
3	2	3	5																																				
2	4	2	4																																				

Q3	Given an image of size 3*3 as $f(m,n)=$ <table><tr><td>128</td><td>212</td><td>255</td></tr><tr><td>54</td><td>62</td><td>124</td></tr><tr><td>140</td><td>152</td><td>156</td></tr></table> Determine the output image $g(m,n)$ using logarithmic transformation by choosing i) $C=1$ ii) $C = \frac{L}{\log_{10}(1+L)}$	128	212	255	54	62	124	140	152	156	10	CO1
128	212	255										
54	62	124										
140	152	156										



Maximum Marks:30		Semester: January 2024- April 2024	
Examination: In-Semester Examination		Duration :1hr 15min	
Programme code:	Class:	Semester:	
Programme: UG BTech	FY/SY/TY/LY /MTECH	I/II/III/IV/V/VI/VII/VIII (SVU 2020/ SVU 2023)	
Name of the Constituent College:		Name of the department:	
K. J. Somaiya College of Engineering		COMP/ETRX/EXTC/IT/MECH	
Course Code: 116U01C601	Name of the Course: Digital Signal & Image Processing		

Ques tion No.		Max. Marks																		
Q1	Perform circular convolution on two sequences $x_1(n)$ and $x_2(n)$ by graphical method only. (periodic sequence.) $x_1(n) = \{0.2, 0.4, 0.6, 0.8, 1.0, 1.2, 1.4, 1.6\}$ $x_2(n) = \{0.1, 0.3, 0.5, 0.7, 0.9, 1.1, 1.3, 1.5\}$	10 M																		
Q2	Perform histogram equalization and draw new equalized histogram for the following image data: <table><tr><td>Grey Level</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>No. of pixels</td><td>700</td><td>1000</td><td>950</td><td>600</td><td>300</td><td>250</td><td>105</td><td>95</td></tr></table>	Grey Level	0	1	2	3	4	5	6	7	No. of pixels	700	1000	950	600	300	250	105	95	10 M
Grey Level	0	1	2	3	4	5	6	7												
No. of pixels	700	1000	950	600	300	250	105	95												
OR																				
Q2	Perform cross correlation of the sequences by graphical method only. $x(n) = \{1, 1, 2, 2\}$ and $y(n) = \{1, 0.5, 1\}$ ↑ ↑	10 M																		
Q3	Justify the statement "Median filter is an effective tool in reducing salt and pepper noise with an example".	10 M																		



Maximum Marks: 30 Programme code: 01 Programme: B.Tech. Computer Engineering		Semester: January 2025-April 2025 Examination: In-Semester Examination	Duration: 1.15 Hrs.
Institute/School/ Department: K. J. Somaiya School of Engineering		Class: TY	Semester: VI (SVU 2020)
Course Code: 116P01C601	Name of the department: COMP		
Name of the Course: DSIP			

Max. Marks: 30

Duration: 1 hr. 15 min

Class: TYBTECH

Semester: VI

Branch: COMP

Course Code: 2UCC60

Name of the Course: Digital Signal and Image Processing

Question No.		Max. Marks												
1	a) Find the signal given below is periodic or aperiodic $y(n) = \sin\left(\frac{2\pi}{7}n^2\right)$ b) Find even and odd components of the given discrete time signal. $x(n) = \{-1, 2, 3, 4, -1, 2, -2, 1, 3\}$ ↑	8M												
2	Test whether following systems are linear or nonlinear a) $y(n) = 2nx(n)$ b) $y(n) = e^{x(n)}$ OR Test following systems for time invariance a) $y(n) = x(n) + x(n+1)$ b) $y(n) = 4nx(n)$	6M												
3	Perform following point processing operations on the digital image 3 bits/pixel image given below. a) Gray level slicing with and without background Given $r_1 = 3$ and $r_2 = 5$ and r must be following the range as $r_1 \leq r \leq r_2$. Consider slicing extreme levels as 0 and 7. b) Bit plane slicing <table border="1" style="margin-left: auto; margin-right: auto;"><tr><td>2</td><td>4</td><td>5</td><td>7</td></tr><tr><td>6</td><td>5</td><td>7</td><td>0</td></tr><tr><td>1</td><td>3</td><td>0</td><td>4</td></tr></table>	2	4	5	7	6	5	7	0	1	3	0	4	8M
2	4	5	7											
6	5	7	0											
1	3	0	4											

4	Explain any two of the following spatial enhancement techniques with suitable example and state one application of each. a) Contrast stretching b) Averaging filter in spatial domain c) Log Transformation	8M
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